

2. Suppose that $\int_1^9 \left(3g(x) + \frac{2}{\sqrt{x}} \right) dx = 29$ and $\int_1^4 g(x) dx = 2$ for some integrable function $g(x)$.
the value of $\int_4^9 g(x) dx$?

- (a) 7
- (b) 27
- (c) 5
- (d) 10/3
- (e) 3

3. Suppose $G(x) = \int_x^0 4g(t) dt$ and $g(x) = \int_0^{3x} \sin(t^2) dt$. What is $G''(x)$?

- (a) $G''(x) = -12 \sin(9x^2)$
- (b) $G''(x) = 12 \cos(9x^2)$
- (c) $G''(x) = 4 \sin(9x^2)$
- (d) $G''(x) = -8x \sin(3x^2)$
- (e) $G''(x) = -8x \cos(x^2)$

3. Suppose $g(x) = \int_1^{3x^2} t \tan^2(2t + 1) dt$. What is $g'(x)$?

- (a) $g'(x) = 36x^3 \tan^2(6x^2 + 1)$
- (b) $g'(x) = 3x^3 \tan^2(6x^2 + 1)$
- (c) $g'(x) = 18x^3 \tan^2(6x^2 + 1)$
- (d) $g'(x) = 6x^3 \tan^2(3x^2 + 1)$
- (e) $g'(x) = x \tan^2(3x^2 + 1)$

1. Evaluate the integral $\int \frac{\sqrt{x^5} + x^2 \sin(3x)}{3x^2} dx$

(a) $\frac{2}{9}x^{\frac{3}{2}} - \frac{1}{9}\cos 3x + C$

(b) $\frac{1}{2}x^{\frac{3}{2}} + \frac{1}{9}\cos 3x + C$

(c) $\frac{2}{9}x^{\frac{3}{2}} - \frac{1}{3}\cos 3x + C$

(d) $\frac{2}{9}x^{\frac{3}{2}} + \frac{1}{3}\cos 3x + C$

(e) $\frac{1}{2}x^{\frac{3}{2}} - \frac{1}{9}\cos 3x + C$

2. Suppose the annual (yearly) CO_2 emissions of a certain city are projected to be

$$E(t) = 160e^{0.04t}$$

metric tons, where t is the number of years after 2010.

(a) (3 marks) What is **the average amount** of CO_2 emissions in the years 2016 to 2020? (Simplify if possible, but your answer may involve e .)

(b) (2 marks) What does $160 + \int_0^{12} E'(t) dt$ represent? (You do not need to compute it.)

(c) (2 marks) What does $\frac{1}{12} \int_0^{12} E'(t) dt$ represent? (You do not need to compute it.)

(d) (3 marks) Suppose now that in a nearby town, **the rate of change** of CO_2 emissions is modelled by $F'(t) = 0.6e^{0.3t}$ metric tons, where t is the number of years after 2010. **Find the net change** in CO_2 emissions in the years from 2010 to 2020. (Simplify if possible, but your answer may involve e .)

4. The limit

$$\lim_{n \rightarrow \infty} \left[\sum_{i=1}^n \left(9 - \left(\frac{5i}{n} \right)^3 \right) \Delta x \right]$$

can be expressed as a definite integral on the interval $[0, 5]$, where the summation inside the brackets is R_n , the Riemann sum using n subintervals and right endpoints as sample points.

(a) (2 marks) The limit above represents an integral of the form $\int_a^b f(x) dx$. What are a , b and $f(x)$ in this example?

(b) (1 mark) What is Δx in this example?

(a) (3 marks) Evaluate the indefinite integral $\int \frac{2\sqrt{\arctan(3x)}}{1+9x^2} dx$.

Question 10

Let $F(x)$ be a function whose derivative $F'(x) = f(x)$ is graphed below:



Assuming that $F(0) = -11.75$, what's the first positive value x for which $F(x) = 0$?

- ☐ 6
- ☐ 6.5
- ☐ 8
- ☐ 5
- ☐ 7.5

Consider the continuous function $g(x)$ defined for all x .

Let $G(x)$ be the antiderivative of $g(x)$ that satisfies $G(8) = -4$.

Then

$$G(x) = \text{[Select]} + \int_a^b \text{[Select]}$$

where

$$a = \text{[Select]}$$

$$b = \text{[Select]}$$